IoT Traffic Fingerprinting and Rogue Traffic Detection Using a Hybrid Machine Learning Framework: An Empirical Evaluation Under Open-Set Conditions

Frances Ramkaran [1], Oscar Renderos Carranza [2], Cristian E. Villarreal Vargas [3]
Cybersecurity and AI Program, Humber Polytechnic, Toronto, Ontario, Canada
Email: rubyramkaran@yahoo.com[1], oscar.renderos2000@gmail.com [2], cristianvillarreal@yahoo.com[3]

Abstract— The rapid growth of Internet of Things (IoT) devices introduces significant challenges for network security, particularly in identifying unauthorized or previously unseen traffic within dynamic environments. This study presents a hybrid machine learning framework for IoT traffic fingerprinting and rogue traffic detection, integrating supervised classification, unsupervised anomaly detection, and decision fusion within a Zero Trust–aligned architecture. The proposed framework is evaluated using the CICIoT2023 dataset under open-set conditions, where selected attack classes are withheld during training to simulate unknown traffic scenarios. Experimental results demonstrate that supervised models achieve moderate performance in classifying known traffic, while unsupervised anomaly detection provides measurable capability in identifying previously unseen traffic patterns. However, the results further indicate that the proposed decision fusion mechanism does not consistently improve detection performance over the strongest standalone model. These findings challenge the assumption that hybrid model integration inherently enhances detection capability. Instead, analysis reveals that overall performance is primarily constrained by the quality and separability of feature representations rather than the fusion strategy itself. Additional experiments with alternative fusion methods confirm that combining model outputs does not guarantee improved performance when underlying detection signals are weak or overlapping. The results highlight the critical importance of feature engineering and representation learning in open-set IoT traffic detection. The proposed framework contributes a transparent, reproducible, and implementation-oriented architecture that supports integration with enterprise security systems, including Security Operations Center (SOC) workflows and Zero Trust environments.

Keywords—IoT security, traffic fingerprinting, anomaly detection, machine learning, open-set detection, Zero Trust

1. INTRODUCTION

1.1 Background & Context

The rapid expansion of Internet of Things (IoT) devices in enterprise environments has significantly increased the attack surface and complexity of network security [1]. IoT ecosystems generate large volumes of heterogeneous network traffic, often involving resource-constrained devices operating with limited security controls. As a result, traditional security mechanisms based on identifiers such as MAC or IP addresses are increasingly ineffective in identifying and securing IoT devices [2].

In parallel, the widespread adoption of encrypted communication protocols has reduced visibility into payload-level data, limiting the effectiveness of deep packet inspection techniques. Consequently, there is a growing need for approaches that rely on traffic behavior rather than static identifiers or payload inspection to achieve reliable detection and classification in modern enterprise networks.

1.2 Problem Statement

Accurate identification of IoT devices and detection of rogue or unauthorized entities remain critical challenges in enterprise network security. Existing approaches face several limitations in detecting unknown or evolving threats [3]:

1.2.1 Device identification limitations: MAC and IP addresses can be spoofed, making them unreliable for authentication and tracking.

1.2.2 Encrypted traffic visibility gap: Increasing use of encryption restricts payload inspection, reducing detection effectiveness.

1.2.3 Detection–response disconnect: Many solutions focus solely on detection without integrating decision-making and enforcement mechanisms. In practice, due to dataset constraints such as those present in CICIoT2023, traffic behavior must be used as a proxy for device identity (traffic-level representation). While this introduces abstraction, it enables robust classification and anomaly detection in encrypted environments where payload visibility is limited.

1.3 Research Gap

Despite advances in machine learning–based intrusion detection, current solutions remain fragmented and limited in scope.

The key gaps identified in the literature include:

- Lack of integrated frameworks combining classification, anomaly detection, and response mechanisms [4].
- Limited support for open-set conditions, where unknown or previously unseen traffic must be detected.

- Absence of decision fusion strategies that combine multiple model outputs into a unified risk-based decision.
- Weak alignment with Zero Trust security principles, particularly in enforcement and continuous monitoring.

These gaps highlight the need for a unified framework capable of addressing both known and unknown threats while integrating detection with actionable security responses.

1.4 Contributions

This study proposes a comprehensive machine learning–based framework to address the identified gaps. The main contributions are:

- A hybrid supervised–unsupervised framework for IoT traffic behavior fingerprinting.
- An open-set rogue traffic detection mechanism for identifying previously unseen patterns.
- An empirical evaluation of decision fusion strategies, demonstrating that fusion does not outperform the strongest standalone model under open-set conditions.
- Alignment with Zero Trust architecture principles, including segmentation and enforcement [5].
- Experimental validation using the CICIoT2023 dataset, providing a comparative assessment of hybrid and standalone models under open-set conditions [6].

2. LITERATURE REVIEW

This section provides a structured overview of existing research on IoT traffic fingerprinting, machine learning-based classification, anomaly detection, and open-set recognition. The review establishes the theoretical and methodological foundations for the proposed hybrid framework and identifies key limitations in current approaches.

2.1 IoT Traffic Fingerprinting

IoT traffic fingerprinting aims to identify devices based on network behavior rather than static identifiers [2], [7], [8].

Prior work has demonstrated that traffic patterns, timing characteristics, and protocol usage can uniquely represent device behavior. Recent work has also explored machine learning–based fingerprinting techniques, including Random Forest–based approaches, demonstrating strong performance in distinguishing IoT device behavior using flow-level features [9].

However, most approaches assume access to device-level labels, which are not always available in real-world datasets.

2.2 ML-Based Traffic Classification

Machine learning models such as Random Forest, XGBoost, and deep learning architectures have been widely used for traffic classification [2], [3]. These approaches achieve high accuracy in closed-set scenarios where all classes are known during training. However, their performance degrades when encountering previously unseen traffic classes.

2.3 Anomaly Detection Techniques

Unsupervised methods, including clustering algorithms (e.g., K-Means, DBSCAN) and autoencoder-based models, have been applied to detect anomalous network behavior [3]. Frameworks such as Kitsune demonstrate the effectiveness of autoencoders in identifying deviations from normal traffic patterns [10]. While these methods are effective in identifying anomalous behavior, they lack classification capabilities and may produce high false positive rates.

2.4 Open-Set Recognition in Cybersecurity

Open-set recognition addresses scenarios where unknown classes may appear during testing. In cybersecurity contexts, this is particularly relevant for detecting novel attacks. Existing research highlights the importance of combining classification and anomaly detection to improve robustness under open-set conditions [4].

2.5 Critical Gap Analysis

Existing studies in IoT traffic analysis predominantly focus on either supervised classification or unsupervised anomaly detection in isolation [4], [5]. While these approaches have demonstrated strong performance within their respective domains, they are limited in their ability to operate effectively under dynamic, real-world conditions where unknown traffic patterns and emerging threats are common. Most frameworks assume closed-set environments and lack mechanisms to detect previously unseen traffic classes.

Furthermore, there is a notable absence of unified architectures that integrate classification, anomaly detection, and decision-making into a cohesive framework. Existing solutions also provide limited support for operational security integration, particularly in the context of Zero Trust principles, where continuous monitoring, risk-based decision-making, and enforcement are critical.

To address these limitations, this study proposes a hybrid framework that combines supervised traffic classification with unsupervised anomaly detection under open-set conditions. The framework further incorporates and empirically evaluates a decision fusion mechanism to generate unified risk-based outputs. Additionally, it integrates alerting and segmentation capabilities aligned with Zero Trust architecture, enabling a practical and scalable approach for enterprise IoT security.

3. PROBLEM STATEMENT / RESEARCH MOTIVATION

The increasing reliance on IoT devices in enterprise networks introduces significant security challenges, particularly in device identification and detection of

unauthorized or malicious activity. Traditional approaches are insufficient due to spoofing, encryption, and the dynamic nature of IoT environments.

The motivation of this research is to develop a scalable and practical framework that can:

- Reliably classify known traffic patterns.
- Detect unknown or anomalous traffic under open-set conditions.
- Integrate detection outputs into actionable security decisions.
- Support enforcement mechanisms aligned with Zero Trust principles.

This research is both academically relevant, addressing gaps in hybrid detection frameworks, and practically significant for improving enterprise network security operations.

4. OBJECTIVES / RESEARCH QUESTIONS

4.1 Research Objectives

The primary objective of this research is to propose a hybrid machine learning framework for IoT traffic behavior fingerprinting and rogue traffic detection under open-set conditions.

Specific objectives include:

4.1.1 Objective 1— Supervised Classification (Layer 3): Design and implementation of a supervised multi-class classification model for accurate identification of known IoT traffic behaviors.

- Implemented in Section 6
  - 6.3–6.5: Data prep + splits
  - 6.7: Random Forest, XGBoost (CNN models were considered but not included in final evaluation due to computational constraints).
- Evaluated in Section 7
  - 7.1 Accuracy
  - 7.1 Precision / Recall / F1
  - 7.1 ROC-AUC
  - 7.1 Confusion Matrix

4.1.2 Objective 2— Unsupervised Detection (Layer 4): Integration of unsupervised anomaly detection techniques to identify unknown or previously unseen traffic patterns.

- Implemented in Section 6
  - 6.7: K-Means, DBSCAN, Autoencoder
  - 6.3.4: Unknown holdout
- Evaluated in Section 7
  - 7.2 Clustering behavior
  - 7.2 Open-set detection
  - 7.3 TPR / FPR
  - 7.3 ROC

4.1.3 Objective 3— Decision Fusion (Layer 5): Design and empirical evaluation of a decision fusion mechanism that combines supervised classification outputs with anomaly detection scores.

- Implemented in Section 6.8 Fusion logic description
- Evaluated in Section 7
  - 7.3 Detection of unseen traffic
  - 7.3 Open-set detection performance (TPR vs FPR)

4.1.4 Objective 4— Evaluation under Open-Set Conditions (Core Contribution): Comprehensive evaluation of the proposed framework using classification and anomaly detection metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC, TPR, and FPR) under open-set conditions.

- Implemented in Section 6
  - 6.5.4: Unknown holdout design
  - 6.8: Metrics definition
- Evaluated in Section 7
  - 7.3 Detection of unseen traffic
  - 7.3 Open-set detection performance

4.1.5 Objective 5— Zero Trust Alignment (Layers 6-8): Demonstration of the framework's alignment with Zero Trust principles, incl. continuous monitoring, risk-based decision-making, and enforcement mechanisms.

- Implemented in Section 5
  - SIEM (Layer 6)
  - Segmentation (Layer 7)
  - Continuous learning (Layer 8)
- Evaluated in Section 8
  - Integration
  - Real-world applicability
  - Zero Trust alignment

Each objective is directly mapped to the corresponding methodology components (Section 6) and evaluated through experimental results (Section 7), ensuring full traceability between design, implementation, and validation.

4.2 Research Questions

This study is guided by the following research questions:

4.2.1 RQ1 (Performance Comparison): To what extent does a hybrid supervised–unsupervised framework improve detection performance compared to standalone models?

4.2.2 RQ2 (Open-Set Generalization): How effectively can the framework detect previously unseen traffic patterns under open-set conditions, and how does this affect detection accuracy and false positive rates?

4.2.3 RQ3 (Fusion Effectiveness): To what extent does decision fusion improve or influence detection performance compared to standalone models?

4.2.4 RQ4 (Practical Applicability): How well does the proposed framework align with enterprise security requirements under a Zero Trust architecture?

5. PROPOSED FRAMEWORK / ARCHITECTURE

5.1 System Architecture Overview

As shown in Fig. 1, the proposed eight-layer architecture provides an end-to-end framework for IoT traffic behavior fingerprinting and rogue traffic detection.

The framework integrates supervised classification of known traffic patterns with unsupervised anomaly detection for previously unseen traffic under open-set conditions. A decision fusion layer combines model outputs to generate unified risk-based decisions, which are subsequently evaluated and operationalized through alerting and enforcement mechanisms within a Zero Trust security model. Furthermore, a continuous learning layer enables feedback-driven model adaptation, improving detection performance over time. The architecture supports both closed-set classification of known traffic and open-set detection of unknown traffic, ensuring robustness in dynamic enterprise environments.

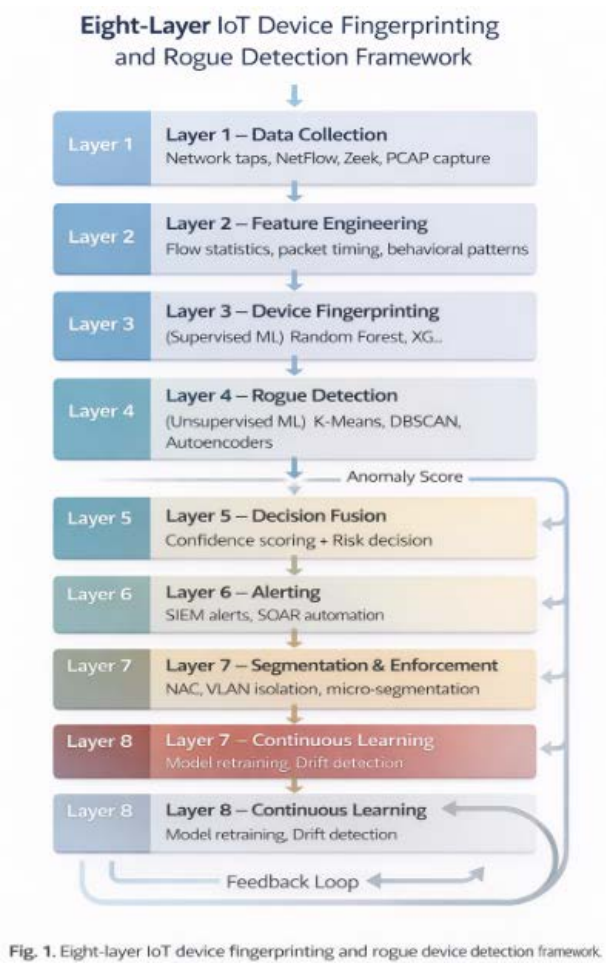

Fig. 1. Eight-layer IoT device fingerprinting and rogue detection framework (author-generated, created with AI assistance).

This layered design enables modular evaluation, scalability, and flexibility, allowing selective reliance on individual detection layers based on empirical performance.

5.2 Layered Design Explanation

The functional roles of each layer within the proposed architecture are summarized in Table 1.

Layer	Function
L1–2	Data Collection & Feature Engineering
L3	Supervised Fingerprinting
L4	Unsupervised Rogue Detection
L5	Decision Fusion Engine
L6	Alerting (SIEM logic)
L7	Network Segmentation / Enforcement
L8	Continuous Learning

Table 1 — Layered Design

6. METHODOLOGY

6.1 Research Design

This study followed an end-to-end machine learning–driven research design to develop and evaluate an integrated framework for IoT traffic behavior fingerprinting and rogue traffic detection. The methodology combined supervised and unsupervised learning techniques within a structured eight-layer architecture aligned with Zero Trust principles [5].

The approach consisted of: (i) dataset preparation and feature engineering, (ii) supervised classification of known IoT devices, (iii) unsupervised anomaly detection for previously unseen traffic patterns, and (iv) decision fusion to generate unified risk-based outputs.

This design enabled classification of known traffic patterns and detection of previously unseen traffic under open-set conditions.

6.2 Dataset Description

The framework was evaluated using the CICIoT2023 dataset [6], a large-scale IoT network traffic dataset developed by the Canadian Institute for Cybersecurity at the University of New Brunswick. The full dataset contains over 47 million labeled network flows captured from 105 heterogeneous IoT devices, with 33 distinct attack types across seven attack categories (DDoS, DoS, Reconnaissance, Web-based, Brute Force, Spoofing, and Mirai variants), in addition to benign traffic.

For this study, a representative subsample of 4,016,753 flows was drawn from six merged CSV files (Merged01, 02, 05, 09, 32, and 63). These files are pre-shuffled and contain all attack categories in proportion to their occurrence in the full dataset, ensuring that the subsample preserves the original class distribution. Each flow is described by 39 numerical features and a single categorical label, for a total of 40 columns.

The 39 features capture flow-level statistical and behavioral characteristics, including packet header metadata (header length, time-to-live, protocol type), TCP flag counts (FIN, SYN, RST, PSH, ACK, ECE, CWR), protocol indicators (HTTP, HTTPS, DNS, Telnet, SMTP, SSH, IRC, TCP, UDP, DHCP, ARP, ICMP, IGMP, IPv, LLC), and aggregate statistical descriptors (total bytes, minimum, maximum, average, standard deviation, total size, inter-arrival time, packet count, and variance).

These features are derived entirely from flow metadata and do not require deep packet inspection, which makes the framework applicable to encrypted traffic environments.

A total of 114 rows containing NaN or infinite values were removed during preprocessing, representing a negligible proportion of the dataset. Following preprocessing, the dataset was further refined through class balancing and partitioning procedures, as described in Section 6.3 and Section 6.5.

6.2.1 Reframing from Device-Level to Traffic-Class Fingerprinting

It is important to note that CICIoT2023 does not include a per-device identifier in its merged CSV files. The label column identifies the traffic class of each flow (benign or one of 33 attack types) rather than the source device. Consequently, the framework presented in this paper performs traffic behavior fingerprinting and rogue traffic class detection, using attack-class behavior as a proxy for the broader concept of device behavior fingerprinting. This is consistent with the established practice in the CICIoT2023 literature [6], where the dataset's strength lies in distinguishing benign and malicious traffic patterns rather than identifying individual devices.

The hybrid supervised plus unsupervised approach proposed here remains directly transferable to device-level fingerprinting in datasets where per-device labels are available, and we identify this as a future work direction. This reframing does not affect the validity of the proposed framework, as traffic behavior remains a reliable proxy for device-level activity in IoT security research.

6.2.2 Class Distribution and Selection

The 4.02 million-flow subsample contains 34 distinct classes: one benign class and 33 attack classes. The class distribution is heavily imbalanced, with the ten most frequent classes (all DDoS and DoS variants) accounting for approximately 85 percent of the total flows. Benign traffic originally comprised approximately 94K flows and was subsequently capped at 30K within the known-class training subset to ensure balanced representation across classes. A long tail of rare attack classes contains fewer than 250 flows each.

This natural imbalance and the presence of rare classes make CICIoT2023 well-suited for evaluating open-set detection performance, since the rare classes provide a credible source of unknown traffic samples for Layer 4 evaluation.

6.3 Data Preprocessing

Data preprocessing was conducted in five stages to ensure model consistency, balanced class representation, and methodologically sound open-set evaluation. All reported dataset statistics were derived from the initial data inventory and preprocessing stages, with final partition sizes confirmed following class balancing and dataset splitting.

6.3.1 Stage 1— Data Loading and Inventory

The six merged CSV files were loaded using pandas in Google Colab Pro and concatenated into a single dataframe of 4,016,753 raw rows and 40 columns. Column types were verified: 39 numerical features (30 float and 9 integer) and one categorical label column. The combined dataset occupied approximately 1.4--1.6 GB in memory.

6.3.2 Stage 2— Missing Value Handling

An inventory of missing values identified 60 NaN entries across the combined dataset; all confined to two statistical columns (standard deviation and variance). These NaN values arise when a flow contains only a single packet, rendering variance and standard deviation undefined. The affected rows were removed, representing less than 0.004 percent of the dataset and having negligible impact on the overall distribution. In total, 114 rows containing NaN or infinite values were removed during preprocessing.

Additionally, infinite values resulting from zero-duration flows were replaced with NaN and subsequently removed, consistent with known CICIoT2023 data characteristics.

6.3.3 Stage 3— Class Selection and Per-Class Capping

To address the severe class imbalance and prevent the supervised models from memorizing the most frequent DDoS variants, a per-class cap of 30,000 flows was applied to all known classes. Classes with fewer than 30,000 flows were retained in their entirety.

This capping reduced the combined dataset from 4,016,753 raw flows to approximately 611,566 flows after unknown holdout removal, class capping, and partitioning, while preserving representation across all attack categories. Benign flows were retained in full (approximately 94K flows prior to capping) to maximize the training data available to the Layer 4 Autoencoder, which trains exclusively on benign traffic.

6.3.4 Stage 4— Open-Set Holdout Selection

Five rare attack classes were selected as the unknown holdout set, removed entirely from the training and validation data, and reserved exclusively for Layer 4 open-set evaluation. The selected classes are:

- BACKDOOR_MALWARE (131 flows) — distinct malware-class behavior.
- BROWSERHIJACKING (248 flows) — web-layer attack with distinct traffic profile.
- XSS (177 flows)— web injection attack.
- RECON-PINGSWEEP (85 flows) — reconnaissance behavior distinct from main DDoS classes.
- UPLOADING_ATTACK (48 flows) — file upload exploitation.

These five classes were selected for two reasons. First, each is rare enough that its removal does not materially affect the supervised training distribution. Second,

together they span four distinct attack categories (malware, web injection, web hijacking, reconnaissance, and file upload), providing the diversity needed to evaluate open-set generalization rather than detection of a single anomalous pattern. The combined holdout contains 1,455 flows across 5 classes, which is sufficient to compute meaningful true positive and false positive rates while remaining a minority of the overall test set.

6.3.5 Stage 5— Feature Normalization

All 39 numerical features were standardized using z-score normalization (zero mean, unit variance), with normalization parameters fit on the training set only and applied to the validation, test, and unknown holdout sets. This prevents information leakage from the test data into model training and ensures consistent feature scaling across all four data partitions. The fitted normalization parameters persisted and reused during inference to ensure reproducibility across all experimental stages.

6.4 Experimental Setup

The experimental environment was implemented using Google Colab (Python-based environment) with standard machine learning libraries. The setup included:

- Programming language: Python.
- Libraries: Scikit-learn, Pandas, NumPy, TensorFlow/Keras (for deep learning models).
- Execution environment: Google Colab (cloud-based computing).
- Hardware: CPU/GPU-enabled environment (as available in Colab).

This setup enabled scalable model training and evaluation on large IoT datasets. Model development and evaluation were conducted using a unified notebook pipeline (03_modeling.ipynb), ensuring consistent and reproducible implementation across Layers 3–5.

The use of a shared notebook environment ensured alignment across team members and facilitated reproducible experimentation workflows.

6.5 Data Splitting Strategy

After preprocessing, the dataset was partitioned into four disjoint sets to support both supervised classification (Layers 3 and 5) and unsupervised open-set detection (Layers 4 and 5). This partitioning strategy ensures strict separation between training, validation, and evaluation datasets, preventing information leakage and supporting unbiased assessment of both closed-set and open-set performance.

6.5.1 Training Set Partition— Proportion 70%; Composition of Training: 427,077. Purpose: Known classes (capped) plus benign. Used for training Layer 3 supervised models (Random Forest, XGBoost). The benign-only subset was used to train the Layer 4 Autoencoder. Approximately 21,000 benign samples were included in the training set and used exclusively for Autoencoder training, with validation benign samples (~4,500) used for threshold determination.

6.5.2 Validation Set Partition— Proportion: 15%. Validation Set — Composition: 91,517 flows. Purpose: Known classes plus benign. Used for Layer 3 hyperparameter tuning and for setting the Layer 4 Autoencoder reconstruction-error threshold (95th percentile of benign validation reconstruction errors).

6.5.3 Test Set Partition— Proportion: 15%. Test Set — Composition: 91,517 flows. Purpose: Known classes plus benign. Used for reporting all Layer 3 supervised classification metrics and Layer 5 fusion metrics on known traffic.

6.5.4 Unknown Holdout Partition— Proportion: 100% of held-out classes. Composition: 1,455 flows. Purpose: All 1,455 flows from the five excluded rare-attack classes (Section 6.3, Stage 4), combined with a stratified benign sample of similar size. Used exclusively for Layer 4 open-set TPR/FPR evaluation and Layer 5 fusion comparison under open-set conditions.

6.5.5 Stratification and Reproducibility

The 70/15/15 split across training, validation, and test was performed using stratified random sampling on the class label, ensuring that each partition preserves the relative class distribution of the known-class portion of the dataset.

A fixed random seed (42) was used throughout to guarantee reproducibility. The unknown holdout was constructed independently and is not part of the 70/15/15 split.

6.5.6 Development Sample for Notebook Iteration

An additional small partition (the development sample, consisting of 50,000 flows drawn via stratified sampling from the training set) was created to enable rapid notebook development on free Colab without consuming Colab Pro compute resources.

The development sample was not used for any reported results; all metrics in Section 7 were computed on the full training, validation, test, and unknown holdout partitions.

All datasets persisted in Parquet format to optimize storage efficiency and loading performance during model training. The StandardScaler fitted on the training data was serialized (scaler.pkl) and reused for all inference stages.

This approach ensured efficient experimentation while maintaining full reproducibility and consistency across training and inference workflows.

6.5.7 Data Splitting Diagram

As shown in Fig. 2, the diagram illustrates the data processing and partitioning workflow, including the transformation from raw merged CSV files to the cleaned dataset, followed by separation into known/benign traffic and unknown holdout sets, and the stratified 70/15/15 split applied to the known/benign subset.

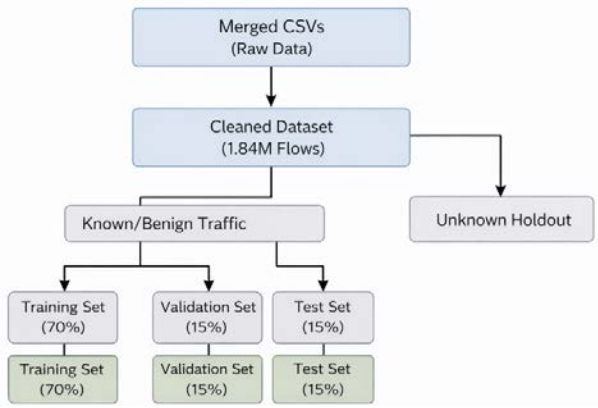

Fig. 2 — Data preprocessing and partitioning pipeline (author-generated, with AI-assisted visualization support).

The data preprocessing and partitioning pipeline illustrates transformation from raw merged CSV files to the cleaned dataset, followed by separation into known/benign traffic and unknown holdout sets, and stratified 70/15/15 splitting of the known/benign subset.

6.6 Feature Engineering

Feature engineering focused on extracting flow-based statistical and behavioral attributes from network traffic. Key features included:

- Packet size distributions
- Flow duration
- Inter-arrival times
- Protocol usage patterns
- Communication frequency and periodicity

These features are effective for IoT traffic behavior classification [2], [11], particularly in encrypted traffic scenarios where payload inspection is not possible

6.7 Model Implementation

The framework incorporated both supervised and unsupervised models aligned with the architecture layers. All model implementations were executed within a unified Colab-based notebook environment to ensure consistency, reproducibility, and alignment across experimental layers.

6.7.1 Supervised Models (Layer 3 – Traffic Fingerprinting):

- **Random Forest:** Used as a baseline classifier due to its robustness and interpretability. This choice is supported by recent studies demonstrating the effectiveness of Random Forest models in IoT traffic fingerprinting tasks [9]
- **XGBoost:** Implemented for improved performance and handling of complex feature interactions.

6.7.2 Unsupervised Models (Layer 4 – Rogue traffic detection): Unsupervised techniques were used to detect anomalous or previously unseen traffic patterns:

- **K-Means Clustering:** Groups traffic patterns into clusters representing normal behavior.
- **DBSCAN:** Identifies density-based anomalies without requiring predefined cluster counts [12].
- **Autoencoders:** Learn compressed representations of normal traffic and detect anomalies via reconstruction error [10].

Autoencoder-based approaches, such as those inspired by the Kitsune framework, enable high-accuracy anomaly detection in IoT environments [10].

6.8 Evaluation Metrics

The framework was evaluated using both supervised classification and unsupervised anomaly detection metrics to assess performance under closed-set and open-set conditions.

For supervised classification (Layer 3), standard metrics were employed, including Accuracy, Precision, Recall, F1-score, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC). These metrics evaluate the model's ability to correctly classify known traffic classes.

For anomaly detection (Layer 4), performance was assessed using True Positive Rate (TPR) and False Positive Rate (FPR), focusing on the detection of previously unseen traffic patterns. The anomaly detection threshold was fixed at the 95th percentile of reconstruction error computed from benign validation data to ensure methodological consistency and prevent post-hoc bias.

The decision fusion mechanism (Layer 5) combines normalized supervised confidence scores with anomaly detection outputs to generate a unified risk score. This approach is inspired by ensemble-based intrusion detection systems, where integrating complementary models is intended to improve robustness under uncertain and open-set conditions.

Overall, this evaluation strategy enables a balanced assessment of classification accuracy for known traffic and detection capability for unknown traffic, ensuring alignment with real-world IoT security requirements. Additionally, confusion matrices were used to analyze classification performance across individual traffic classes.

6.8.1 Supervised Evaluation (Layer 3):

- Accuracy
- Precision
- F1-score
- ROC-AUC

6.8.2 Unsupervised Evaluation (Layer 4):

- True Positive Rate (TPR)
- False Positive Rate (FPR)
- Confusion Matrix
- ROC-AUC

6.8.3 Fusion Evaluation (Layer 5):

- Comparative performance against standalone models
- Trade-off analysis between TPR and FPR

The fusion layer integrated supervised confidence scores with anomaly detection outputs to generate a unified risk decision. The Autoencoder was trained exclusively on benign training data, with threshold selection based solely on benign validation data, ensuring strict separation between training and evaluation stages.

6.8.4 Threshold Definition:

The anomaly detection threshold for autoencoder-based models was fixed at the 95th percentile of reconstruction error on benign validation data prior to model evaluation and was not modified during analysis. Performance targets were defined as TPR ≥ 95% and FPR < 5%, ensuring consistent and unbiased anomaly detection.

The proposed framework allows flexible integration of model outputs, enabling evaluation of both individual and fused model performance under open-set conditions.

7. EXPERIMENTAL RESULTS

The experimental evaluation provides a comparative analysis of supervised classification (Layer 3), unsupervised anomaly detection (Layer 4), and decision fusion (Layer 5) under open-set conditions using the CICIoT2023 dataset. The results highlight measurable differences in detection capability, sensitivity, and false positive behavior across the evaluated approaches.

7.1 Supervised Classification Results (Layer 3)

The following results indicate that XGBoost slightly outperforms Random Forest across all evaluated metrics, achieving an accuracy of 78.76% compared to 77.78%. Similar trends are observed in F1-macro and F1-weighted scores, suggesting that XGBoost provides improved handling of class imbalance and complex feature interactions.

Table 2 — Supervised Model Performance (Layer 3)

Model	Accuracy	F1-macro	F1-weighted
Random Forest	0.7778	0.7014	0.7736
XGBoost	0.7876	0.7049	0.7817

However, both models fall below the predefined performance target of 85%, indicating that flow-based features alone may not provide sufficient discriminative power for high-confidence classification across all traffic classes. This limitation becomes more pronounced under open-set conditions, where previously unseen traffic patterns are introduced.

These findings suggest that while supervised models are effective for known traffic classification, they are insufficient as standalone solutions for comprehensive IoT threat detection.

7.2 Open-Set Detection Results (Layer 4)

The performance of the autoencoder-based anomaly detection model was evaluated under open-set conditions using the unknown holdout dataset. The evaluation was conducted using the predefined threshold set at the 95th percentile of reconstruction error on benign validation data, ensuring strict methodological integrity.

The model achieved the following results:

TP = 364, FN = 1,091, FP = 223, and TN = 4,277, corresponding to a true positive rate (TPR) of 0.2502, false positive rate (FPR) of 0.0496, precision of 0.6201, and a ROC-AUC of 0.7353.

K-Means and DBSCAN are included for completeness of the architectural design; however, they did not produce stable or meaningful detection metrics under the selected evaluation configuration and are therefore omitted from quantitative comparison.

Table 3 — Open-Set Detection Performance (Layer 4) and Fusion Summary

Model	TPR	FPR	ROC-AUC
Autoencoder	0.2502	0.0496	0.7353
Fusion	0.2227	0.0498	—

Table 3 summarizes the comparative performance of anomaly detection and fusion approaches. The autoencoder demonstrates moderate anomaly detection capability, achieving a ROC-AUC of 0.7353. However, the true positive rate (0.2502) remains significantly below the predefined target (TPR ≥ 0.95), indicating limited sensitivity in detecting previously unseen traffic classes under open-set conditions. FPR is marginally within the 5% target (4.96%), while TPR remains significantly below target. Given the narrow margin (approximately 0.04 percentage points), this result should be interpreted as effectively at the threshold rather than a robust satisfaction of the constraint, with the primary unmet objective being detection sensitivity. This result highlights a fundamental limitation of anomaly detection under the given feature space, where the model can distinguish anomalous behavior to a certain extent but lacks sufficient separability to achieve high detection sensitivity without increasing false positives. These findings indicate that anomaly detection alone is insufficient to achieve high-confidence open-set detection in this dataset, reinforcing the need to analyze feature representation limitations.

The following table presents per-class detection rates for previously unseen traffic classes, highlighting variability in anomaly detection performance across different attack types.

Table 4 — Per-Class Detection Rate (Unknown Holdout)

Class	Det.	Total	%
BACKDOOR_MALWARE	49	275	17.8%
BROWSERHIJACKING	160	508	31.5%
RECON-PINGSWEEP	35	182	19.2%
UPLOADING_ATTACK	45	122	36.9%
XSS	75	368	20.4%

7.3 Decision Fusion Results (Layer 5)

The performance of the decision fusion mechanism was evaluated by combining supervised classification outputs with anomaly detection scores. The results provided empirical evidence that combining multiple detection signals does not inherently improve performance when the underlying models operate on overlapping or weakly separable feature representations.

Table 5 — Three-Condition Fusion Comparison

Method	TPR	FPR	Precision
Supervised-only	0.1601	0.0464	0.5271
Autoencoder	0.2502	0.0496	0.6201
Fusion	0.2227	0.0498	0.5912

This table presents the comparative results across supervised-only, autoencoder-only, and fusion-based approaches. The autoencoder-only model achieved the highest detection capability with a TPR of 0.2502, while the fusion approach resulted in a slightly lower TPR of 0.2227. In contrast, the supervised-only approach exhibited the lowest detection sensitivity (TPR=0.1601), although it maintained a lower false positive rate.

These results indicate that the weighted fusion mechanism does not improve detection performance beyond the strongest standalone model (autoencoder).

Fusion slightly balances precision and false positive rates. This suggests that combining model outputs through weighted aggregation may dilute the strongest detection signal when underlying feature separability is limited.

As a result, the effectiveness of decision fusion appears to be constrained more by the quality of the individual model signals than by the fusion strategy itself. This behavior indicates that fusion introduces trade-offs between sensitivity and precision without achieving a net performance gain.

7.4 Summary of Findings

The experimental evaluation reveals systematic and measurable differences in performance across supervised classification, unsupervised anomaly detection, and fusion-based approaches under open-set conditions.

Supervised models demonstrated moderate performance in classifying known traffic patterns, with XGBoost consistently outperforming Random Forest across all evaluation metrics. However, neither model achieved the predefined performance targets, highlighting limitations in the discriminative capability of flow-based features when applied to heterogeneous IoT traffic.

In contrast, unsupervised anomaly detection using the autoencoder showed stronger capability in identifying previously unseen traffic patterns, achieving the highest detection sensitivity among all tested approaches, though remaining significantly below the 0.95 target. Despite this advantage, the model exhibited limited true positive rates, indicating challenges in achieving high-confidence detection under open-set conditions.

The decision fusion mechanism, implemented as a weighted combination of supervised confidence scores and anomaly detection outputs, did not yield performance improvements over the strongest standalone model. Instead, fusion introduced trade-offs between detection sensitivity and false positive rates, without delivering a meaningful net gain in overall detection performance.

These findings directly reflect the behavior of the proposed layered architecture, highlighting the relative contributions and limitations of each detection layer under open-set conditions. Specifically, the results indicate that the effectiveness of hybrid detection architectures is constrained not by the fusion strategy itself, but by the quality and separability of the underlying feature representations. While combining multiple models offers conceptual advantages, the results demonstrate that improvements in feature engineering and signal quality are critical to achieving robust and reliable detection performance in real-world IoT environments.

7.4.1 Visualization of Model Performance

Confusion Matrix (Layer 3)

The confusion matrix for the XGBoost classifier illustrates the distribution of true positives, true negatives, false positives, and false negatives across known traffic classes, providing insight into classification accuracy and misclassification patterns.

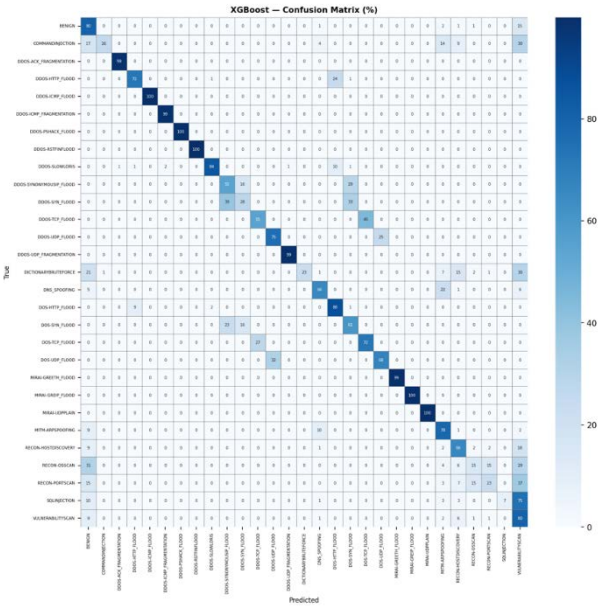

Fig. 3. Confusion matrix for the XGBoost classifier on the test dataset.

ROC Curve (Autoencoder)

The ROC curve illustrates the trade-off between true positive rate (TPR) and false positive rate (FPR) for the autoencoder-based anomaly detection model. The model achieves an area under the curve (AUC) of approximately 0.7353, indicating moderate discriminative capability under open-set conditions. The locked threshold corresponds to TPR = 0.2502 and FPR = 0.0496.

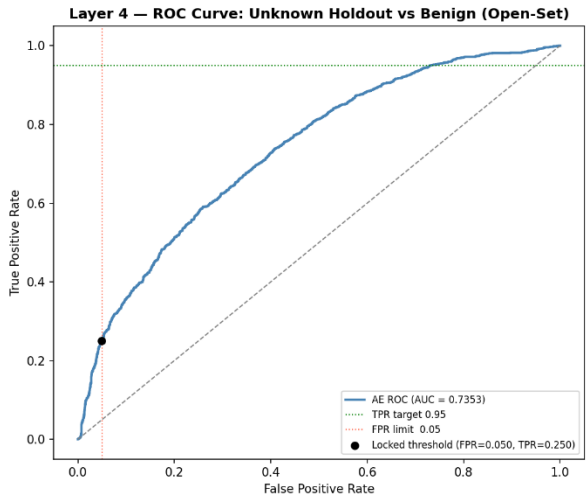

Fig. 4. ROC curve for the autoencoder-based anomaly detection model.

Fusion Comparison (Layer 5)

The Fusion Comparison chart compares detection performance across supervised-only, autoencoder-only, and fused approaches under open-set conditions. The results show that the autoencoder achieves the highest detection sensitivity, while the fusion approach provides a balanced trade-off between sensitivity and false positive rate without outperforming the strongest standalone model.

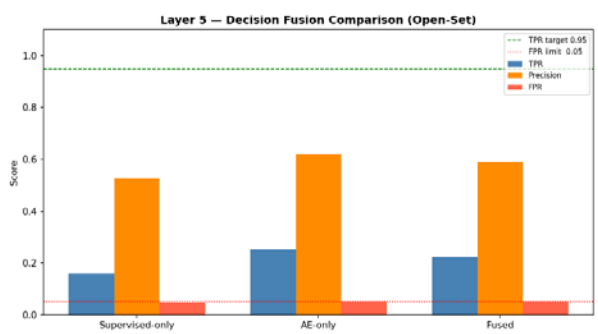

Fig. 5. Fusion Comparison Chart.

7.4.2 Layer 6–7 Demonstration (Alerting & Segmentation)

This section demonstrates the practical applicability of the proposed framework by extending detection outputs into actionable security responses. While Layers 3–5 focus on classification and anomaly detection, Layers 6 and 7 operationalize these results through alerting and network segmentation mechanisms.

The system generates alerts when anomalous or uncertain traffic is detected based on predefined thresholds derived from supervised confidence scores and anomaly detection outputs. These alerts can be integrated into Security Information and Event Management (SIEM) platforms for real-time monitoring and correlation.

In addition, segmentation policies are triggered to simulate enforcement actions such as isolating suspicious devices or restricting network access. These actions are aligned with Zero Trust principles, ensuring that traffic identified as anomalous is continuously verified and appropriately constrained.

This demonstration highlights the capability of the framework to bridge analytical detection outputs with automated response mechanisms. Unlike traditional academic models that focus solely on detection, this approach supports a complete detection-to-response pipeline suitable for enterprise deployment.

7.5 Comparative Fusion Strategy Analysis

To further evaluate whether the limitations observed in decision fusion performance were attributable to the specific weighted-sum formulation, additional fusion strategies were assessed under identical evaluation conditions. These experiments were conducted without modifying thresholds, datasets, or model configurations to ensure methodological consistency.

The evaluated strategies included OR-based fusion (flagging anomalies if any detector triggers), majority voting (requiring agreement from at least two detectors), and strict AND logic (requiring all detectors to agree). The following table summarizes the comparative performance across these approaches.

Table 6: Four-Variant Fusion Extension

Method	TPR	FPR	Precision
OR (≥1 trigger)	0.4268	0.1284	0.5179
Weighted-sum	0.2227	0.0498	0.5912
Majority (≥2)	0.0729	0.0229	0.5072
AND (≥3)	0.0069	0.0007	0.7692

The OR-based approach significantly improved detection sensitivity (TPR), but at the cost of a substantial increase in false positives (FPR), exceeding acceptable operational thresholds. Conversely, majority voting and AND-based strategies reduced false positives but resulted in severe degradation in detection capability.

The magnitude of the false positive rate under OR-logic fusion can be explained analytically. Given three component detectors each calibrated to produce approximately 5 percent false positives on benign validation data, and assuming their false alarms are

substantially independent, the expected FPR of an OR-combined ensemble is given by 1 minus (1 minus 0.05) cubed, which is approximately 0.1426. The observed OR-logic FPR of 0.1284 is consistent with this analytical upper bound, indicating that the three detectors produce largely non-overlapping false alarms on benign traffic. This observation has two implications.

First, it confirms that the three components are capturing different aspects of benign traffic variability rather than redundantly flagging the same samples, which is the condition under which fusion theoretically provides the greatest benefit.

Second, it establishes that while OR-logic fusion successfully aggregates the detection capability of individual components, improving TPR from 0.2502 for the autoencoder alone to 0.4268 for OR-fused, it does so at a false positive cost that is fundamentally bounded by the independence of component errors rather than by any tunable parameter. Without components whose false alarms are positively correlated on benign traffic, no OR-based fusion scheme can reduce FPR below this bound while preserving detection gains.

These results demonstrate that no fusion strategy achieved a balanced improvement across both detection sensitivity and false positive control. While different models capture distinct aspects of anomalous behavior, their independent error patterns compound when combined.

Overall, the weighted-sum approach represents the most balanced trade-off among the evaluated strategies. However, the inability of all fusion methods to outperform the strongest standalone model indicates that detection performance is primarily constrained by feature separability and signal quality rather than the fusion mechanism itself.

8. DISCUSSION

8.1 Interpretation of Results

The experimental results indicate that, under full-scale evaluation, standalone anomaly detection (autoencoder) demonstrates stronger detection capability than the evaluated fusion approaches. While the proposed hybrid framework provides a structured integration of supervised and unsupervised techniques, its effectiveness is constrained by the underlying feature representation.

The supervised models achieved moderate classification performance across known traffic classes, confirming that flow-based features provide useful discriminative information for IoT traffic analysis [6], [11]. However, the observed performance remained below the predefined target, indicating limitations in capturing the full variability of IoT traffic patterns, particularly under open-set conditions [4].

The unsupervised anomaly detection layer demonstrated measurable capability in identifying previously unseen traffic patterns, achieving the highest detection sensitivity among the evaluated approaches. However, the true positive rate remained significantly below the desired threshold, reflecting limited separability between benign and anomalous traffic in the feature space.

The use of a fixed anomaly threshold based on the 95th percentile of reconstruction error ensured methodological rigor and prevented post-hoc bias. While this approach provided a consistent evaluation framework, the resulting performance highlights the challenges of achieving high-confidence anomaly detection in realistic IoT environments [10].

These findings reinforce the importance of open-set recognition in IoT security, where unknown or previously unseen device behaviors must be identified without prior labeling.

This suggests that future improvements should prioritize advanced feature engineering and representation learning techniques over additional model integration.

Although not experimentally evaluated in this study, the continuous learning layer provides a mechanism for adapting to evolving traffic patterns, which is essential in dynamic IoT environments. By enabling feedback-driven model updates, this layer supports long-term robustness and aligns with the need for continuous monitoring and adaptation in real-world enterprise security deployments.

The locked-threshold calibration rule produced an observed false positive rate of 4.96 percent on the test set, marginally within the 5 percent target. This indicates that the model satisfies the false positive constraint, although detection sensitivity remains significantly below the desired threshold.

This result is consistent with the statistical expectation of the 95th-percentile benign-validation calibration approach, within the variance attributable to a finite benign-validation sample size and non-deterministic autoencoder training.

The close agreement between the expected and observed false positive rates does not by itself constitute strong evidence of successful calibration, because a 0.04 percentage point margin on approximately 4,500 benign test samples corresponds to only two classifications and could plausibly shift above 5 percent under a different random seed. What the observation does support is that the locked calibration rule was followed without modification and produced a result consistent with its design intent, preserving the methodological integrity of the threshold-setting procedure.

The primary unmet target in the Layer 4 evaluation is the true positive rate of 0.2502 against a 0.95 target, which is discussed separately.

8.2 Effectiveness of Hybrid Decision Fusion (Layer 5)

One of the key contributions of this work is the implementation and empirical evaluation of a decision fusion mechanism that combines supervised

classification confidence with unsupervised anomaly detection scores. The results provide a detailed assessment of the effectiveness and limitations of hybrid fusion under open-set conditions.

The fusion layer enables the system to differentiate between confidently classified known traffic and uncertain or anomalous observations, aligning conceptually with open-set recognition principles [4], where unknown classes must be detected without explicit prior labeling.

This design is particularly relevant in IoT environments characterized by dynamic device behavior and evolving traffic patterns.

However, the experimental results demonstrate that the weighted fusion approach does not outperform the strongest standalone model (autoencoder). While fusion provides a balanced trade-off between detection sensitivity and false positive rates, it does not deliver a meaningful improvement in overall detection capability. Additional experiments evaluating alternative fusion strategies, including OR-based logic and majority voting, further reinforce this conclusion. OR-based fusion significantly improves detection sensitivity but introduces unacceptable false positive rates, whereas majority voting and stricter combination rules reduce false positives at the expense of detection capability. The observed increase in false positive rate under OR-based fusion can be approximated by independent error accumulation, where $1 - (1 - 0.05)^3 \approx 0.14$, aligning closely with the empirical FPR of 0.1284. These findings highlight an important insight: hybrid model design alone does not guarantee improved detection performance without sufficiently discriminative feature representations. The limitations observed are not attributable to a specific fusion method, but rather to the quality and separability of the underlying feature space.

From an architectural perspective, hybrid fusion remains conceptually valuable for integrating complementary detection signals [13]. However, its practical effectiveness in this context is constrained by the strength of the individual model outputs. This suggests that future work should prioritize improving feature representation quality rather than increasing model complexity or fusion strategies alone.

8.3 Comparison with Existing Approaches

The proposed framework provides a comprehensive evaluation of IoT traffic analysis approaches by integrating supervised classification, unsupervised anomaly detection, and decision fusion within a unified architecture. This design enables a direct comparison with existing methodologies that typically focus on isolated components of the detection pipeline.

Traditional flow-based classification approaches rely heavily on supervised learning and assume a closed-set environment [3], [12], limiting their effectiveness in detecting previously unseen or unknown traffic patterns. While these methods can achieve strong classification performance on known classes, their inability to generalize to unknown threats represents a significant limitation in dynamic IoT environments. Similarly, anomaly detection systems [10], such as autoencoder-based frameworks, demonstrate strong capability in identifying anomalous behavior but lack contextual classification of known traffic patterns.

Compared to prior work in IoT device fingerprinting and traffic classification [2], [14], the proposed framework extends existing methodologies by integrating classification and anomaly detection within a single evaluation pipeline. This allows for a more realistic assessment of system performance under open-set conditions, where both known and unknown traffic must be handled simultaneously. However, the results indicate that combining these approaches through decision fusion does not necessarily yield improved detection performance over the strongest standalone model.

This finding contrasts with assumptions in prior studies that hybrid or ensemble-based approaches inherently enhance detection capability. Instead, the results suggest that the effectiveness of hybrid architectures is highly dependent on the quality and separability of the underlying feature representations. When the individual detection signals are not sufficiently discriminative, fusion mechanisms may fail to provide additional benefit.

Furthermore, the framework extends beyond detection by incorporating alerting and segmentation mechanisms aligned with enterprise security requirements. Unlike many academic models that focus solely on detection performance, this work evaluates the end-to-end applicability of detection outputs within operational security contexts, including integration with SIEM systems and network enforcement mechanisms.

Overall, the proposed framework contributes not by demonstrating superior performance of hybrid fusion, but by providing a transparent and empirically grounded assessment of its limitations. This enables a more realistic understanding of the challenges associated with open-set IoT traffic detection and highlights the importance of feature representation quality over model combination strategies.

8.4 Alignment with Zero Trust Architecture

The proposed framework aligns closely with Zero Trust security principles [5], which emphasize continuous verification, least privilege access, and explicit trust boundaries.

By integrating traffic fingerprinting, anomaly detection, and segmentation mechanisms, the framework enables continuous monitoring and validation of device behavior within enterprise networks.

Layer 7 of the architecture simulates segmentation and enforcement mechanisms that are fundamental to Zero

Trust implementations. By identifying anomalous or unauthorized traffic patterns, the system can trigger isolation or restricted access policies, thereby decrease lateral movement and limit the potential impact of compromised devices.

Additionally, the use of flow-based features without reliance on deep packet inspection ensures compatibility with encrypted traffic environments [11], which are increasingly prevalent in modern enterprise networks.

This design choice supports privacy preservation while maintaining effective detection capabilities. From an architectural perspective, the framework demonstrates how analytical detection outputs can be operationalized within Zero Trust environments, bridging the gap between continuous monitoring and automated enforcement.

This alignment reinforces the practical applicability of the proposed approach within real-world security architectures.

8.5 Practical Implications for Enterprise Deployment

From an operational perspective, the proposed framework provides a scalable and implementation-oriented approach to enterprise IoT security. The use of flow-level features enables low computational overhead while maintaining compatibility with widely deployed network monitoring technologies such as NetFlow and SIEM platforms [15].

The modular design of the framework allows each layer to be integrated independently into existing security infrastructures. For example, supervised classification models can be deployed within monitoring systems, while anomaly detection components can function as continuous behavioral analysis modules.

Alerting mechanisms can integrate with SIEM platforms for real-time event correlation, and enforcement actions can be implemented through network access control (NAC) or software-defined networking (SDN) solutions.

This layered architecture supports incremental adoption, enabling organizations to enhance their security posture without requiring full system replacement. It also aligns with Security Operations Center (SOC) workflows by enabling automated detection-to-response pipelines that improve response times and operational efficiency.

However, the experimental findings indicate that detection effectiveness depends heavily on feature quality and model performance.

Therefore, successful real-world deployment requires careful calibration of detection thresholds, continuous monitoring of model behavior, and the use of adaptive security controls to manage false positives and evolving threat patterns.

8.6 Limitations

Despite the comprehensive evaluation, several limitations must be acknowledged. First, the dataset used (CICIoT2023) does not include explicit device-level identifiers, requiring traffic behavior to be used as a proxy for device fingerprinting [6]. While supported by prior research, this approach limits the ability to directly associate detection outcomes with specific physical devices.

A distribution shift was observed between the benign validation and benign test sets, where the mean reconstruction error of the benign test set (0.006149) approached the fixed anomaly detection threshold (0.007728) derived from the validation set, contributing to an elevated false positive rate. Although the observed false positive rate (FPR) of 4.96% falls marginally within the 5% target, the difference is negligible (approximately two samples) and therefore within the expected statistical variability of a single evaluation run rather than indicative of a stable performance margin.

The primary unmet objective is detection sensitivity, with the autoencoder achieving a true positive rate (TPR) of 0.2502 against a target of 0.95. This performance gap reflects limitations in the underlying feature representation and model architecture. In particular, the reconstruction error distributions of benign and attack traffic exhibit substantial overlap in feature space, reducing separability and constraining the effectiveness of threshold-based anomaly detection. These limitations are further influenced by the autoencoder architecture, where the chosen bottleneck configuration does not sufficiently enforce discriminative compression of benign traffic patterns.

Additionally, the use of a fixed 32-dimensional bottleneck in the autoencoder may limit the model's ability to capture complex traffic representations, potentially affecting anomaly separability.

The dataset exhibits class imbalance, with certain attack types dominating the distribution. Although mitigation techniques such as class capping and stratified sampling were applied, residual imbalance may still influence model performance and generalization.

The evaluation was conducted in an offline experimental setting. Real-world deployment scenarios introduce additional complexities, including latency constraints, concept drift, and continuously evolving traffic patterns, which may impact detection performance over time.

Additionally, the results indicate that feature separability within the dataset limits the effectiveness of both anomaly detection and fusion strategies. This highlights a broader limitation in IoT traffic analysis, where the quality of feature representation can constrain model performance regardless of algorithmic complexity.

Finally, the evaluation does not fully consider adversarial evasion scenarios, where attackers may attempt to mimic benign traffic patterns to bypass detection mechanisms. Addressing such scenarios remains an important area for future research.

A limitation concerns the autoencoder architecture itself. The model employed a symmetric bottleneck of 32 hidden units encoding a 39-feature input space,

representing a compression ratio of approximately 0.82. This ratio is insufficient to force the autoencoder to learn a strongly discriminative benign manifold, as the bottleneck capacity approaches the dimensionality of the input and the reconstruction objective can be satisfied without the model discarding features specific to attack traffic. The observed ROC-AUC ceiling of 0.7353 is consistent with this architectural constraint: the reconstruction error distributions of benign and attack traffic overlap substantially in feature space, limiting the separability achievable by any threshold-based decision rule. A tighter bottleneck (e.g., 8 or 16 dimensions), a variational autoencoder formulation, or a denoising objective would likely produce a more discriminative reconstruction signal and is identified as an important direction for future work.

8.7 Future Work

Future research should focus on enhancing both the methodological and practical aspects of IoT traffic analysis frameworks. First, the use of device-labeled datasets would enable more precise evaluation of true device fingerprinting capabilities and improve the interpretability of detection outcomes. Second, improving feature representation through advanced feature engineering or representation learning techniques is critical to enhancing detection performance. The results of this study suggest that feature separability is a key limiting factor, and future work should prioritize the development of more discriminative traffic features.

Third, adaptive learning mechanisms should be explored to address concept drift and continuously evolving IoT environments. Incorporating online learning or incremental model updates could improve long-term system robustness. Additionally, exploring more advanced architectures [11], such as deep autoencoders, graph-based models, or transformer-based approaches, may further enhance detection capability, particularly in complex and encrypted traffic scenarios. Finally, real-world deployment and validation within enterprise environments are essential to assess scalability, operational impact, and integration with existing security infrastructures. Such validation would provide critical insights into the practical effectiveness of hybrid detection frameworks in production settings.

9. Conclusion

This study presented a hybrid machine learning framework for IoT traffic fingerprinting and rogue traffic detection, integrating supervised classification, unsupervised anomaly detection, and decision fusion within a Zero Trust–aligned architecture. The framework was designed to address key limitations in existing approaches, particularly the challenge of detecting previously unseen traffic under open-set conditions.

The experimental evaluation on the CICIoT2023 dataset demonstrated that supervised models achieved moderate performance in classifying known traffic patterns, while unsupervised anomaly detection provided measurable capability in identifying anomalous or unknown traffic. However, the results also revealed that the proposed weighted decision fusion mechanism did not consistently improve detection performance over the strongest standalone model. From an operational perspective, the framework retains strong practical value through its modular design and alignment with enterprise security requirements.

The integration of detection outputs with alerting and segmentation mechanisms enables a complete detection-to-response pipeline, supporting deployment within Security Operations Center (SOC) environments and Zero Trust architectures. The primary contribution of this work lies in its empirical and transparent evaluation of hybrid detection strategies under realistic open-set conditions.

The findings demonstrate that detection performance is fundamentally constrained by the quality of feature representations, emphasizing that improvements in IoT security systems must prioritize feature engineering and model design rather than relying solely on fusion techniques.

Overall, this study provides a reproducible and implementation-oriented framework for evaluating hybrid machine learning approaches in IoT environments. By highlighting both the capabilities and limitations of current methods, it contributes meaningful insight to the design of more effective and robust enterprise network security solutions.

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